Code

main.py +

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# The parameters BD,BH,Distance are in metres

BD=float(input("Blasting Diameter="))
a=float(input("Spacing to burden ratio="))
p=float(input("Average in-hole density=")) # measured in g/cc
BH=float(input("Bench height="))
Distance=float(input("distance="))
PFt=float(input("PF tech="))        # measured in kg/m**3
rock_type=input("Rock type:")
A=float(input("Rock factor:"))
Explosive_type=input("Type of explosive:")  #Either ANFO or TNT
E=float(input("Relative explosive energy:"))

if Distance > 500:
    SL=20 * BD
else:
    SL=30 * BD
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print("Stemming length is", SL,"m")

M=(p*((BD*1000)**2))/1273 #Multiply by 1000 to convert from metres to millimetres
B=((M*(BH-SL))/(a*PFt*BH))**0.5
print("Burden is",B, "m")
if rock_type=="soft":
   SD=-(1/4*B)
elif rock_type=="hard":
    SD=1/3*B
S=a*B
PFa=(M*(BH-SL+SD))/(B*S*BH)
print("Actual powder factor is", PFa, "kg/m**3")

CL=BH-SL
Q=CL*M
X50=A*(PFt**-0.8)*(Q**0.167)*((115/E)**0.633)
if Explosive_type=="ANFO":
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    E=100
elif Explosive_type=="TNT":
    E=115
print("Mean fragmentation size is", X50, "cm")
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